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Comprehensive Characterization of Zinc-Based H₂S Scavengers Using a Multi-Faceted Approach: Enhancing Safety and Efficiency in Drilling Operations

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Abstract

Hydrogen sulfide (H₂S) scavengers play a critical role in ensuring the safety and efficiency of drilling operations. As the demand for reliable and effective scavengers continues to grow, there is a pressing need for robust methodologies to evaluate their purity and performance. This study addresses this gap by introducing a novel approach to assess the quality of zinc-based H₂S scavengers. Method: A novel multi-faceted approach was employed to assess the purity and performance of zinc-based H₂S scavengers. The methodology involved combining evaporation, Nuclear Magnetic Resonance (NMR) spectroscopy, scavenging capability evaluations, measurement of flash point, and determination of specific gravity to comprehensively characterize the scavengers. This integrated approach enabled the detection of impurities and defects that may compromise scavenger effectiveness. Additionally, the use of NMR allowed for the elucidation of molecular structures, while evaporation studies assessed thermal stability and decomposition patterns. Measurement of flash point ensured the assessment of flammability risk, whereas specific gravity determinations evaluated density variations. Results: The results showed that the combined evaluation methodology successfully identified variations in scavenger purity and performance. The NMR analysis revealed subtle differences in molecular structures, indicating slight deviations in synthesis conditions. Flash point studies highlighted discrepancies in thermal stability, evaluating the potential issues during transportation and storage. Scavenging capability assessments demonstrated strong correlations between scavenger composition and H₂S removal efficiency, underscoring the importance of precise formulation. Moreover, measurements of flash point and specific gravity indicated the overall quality of the chemical, emphasizing the need for careful material selection. Overall, the findings underscored the significance of rigorous testing protocols in ensuring the reliability and quality of zinc-based H₂S scavengers for a safe drilling operation. This study presents a pioneering approach to evaluating the quality of zinc-based H₂S scavengers, setting a new standard for the industry. By integrating multiple analytical techniques, this methodology enables the identification of previously undetectable impurities and defects, thus enhancing the overall safety and efficiency of drilling operations.

Introduction

Hydrogen sulfide (H₂S) is a colorless, highly toxic gas that poses significant hazards in drilling operations. Known for its characteristic rotten egg odor, H₂S can lead to severe health risks for personnel, equipment failure, and environmental contamination. The presence of H₂S in natural gas and crude oil reservoirs necessitates effective management strategies to mitigate risks associated with its release during drilling and production activities. As the oil and gas industry expands into deeper and more challenging environments, the demand for effective H₂S scavengers has intensified.

Among various scavenging agents, zinc-based compounds have emerged as prominent choices due to their efficiency in neutralizing H₂S through chemical reactions that convert the gas into less harmful products. However, the effectiveness of these scavengers can be compromised by impurities and defects arising from manufacturing processes, which may affect their performance and safety (White & Thompson, 2019). The industry has recognized the critical need for robust methodologies to evaluate the purity and performance of these scavengers to ensure operational safety and efficiency.

The objective of this study is to introduce a multi-faceted approach for the comprehensive characterization of zinc-based H₂S scavengers. This research aims to evaluate the purity and performance efficacy of various scavenger formulations using a combination of analytical techniques. By employing methodologies such as Nuclear Magnetic Resonance (NMR) spectroscopy, evaporation studies, flash point measurements, and specific gravity determinations, this study seeks to establish robust protocols for evaluating the quality of scavengers. The anticipated results are expected to provide critical insights into the molecular structure and performance capabilities of these materials, ultimately enhancing safety standards in drilling operations.

Method

Materials

The zinc-based H₂S scavengers used in this study were sourced from various reputable suppliers to ensure a diverse representation of commercially available products. Each sample underwent extensive quality control checks upon receipt to confirm that they adhered to industry standards. The samples were carefully stored under controlled conditions to prevent contamination and degradation, following best practices outlined in previous research. The selected scavengers were chosen based on their popularity in the industry and their varying formulations, which included different concentrations of zinc and other components.

Analytical Techniques

Evaporation Studies. Evaporation studies were conducted to evaporate the base fluid (carrying fluid) of the scavengers. Samples were subjected to evaporation process at varying temperatures, and the rates of evaporation were monitored over time based on the decrease of the initial weight. The results from these studies provide significant understanding about the weight% of each component in the solution since it is composed of two main chemicals. The final weight% will be calculated based on the component's density. In addition, to confirm the zinc-based component percentage, the remaining residue is subjected to NMR with original sample to validate that all solvents have been evaporated.

Nuclear Magnetic Resonance (NMR) Spectroscopy. NMR spectroscopy was employed to elucidate the molecular structures of the zinc-based scavengers. This technique provided insights into the purity of the samples and identified potential deviations in synthesis conditions. NMR is effective in detecting subtle changes in molecular structures that can affect scavenging efficiency. The analysis was performed using a high-field NMR spectrometer, and the resulting spectra were analyzed to identify functional groups and molecular conformations that are critical for interaction with H₂S.

Scavenging Capability Evaluations. The H₂S removal efficiency of each scavenger was quantitatively assessed using standard gas chromatography techniques. This involved passing a known concentration of H₂S gas through a fixed volume of scavenger and measuring the residual concentration of H₂S in the gas phase. The correlation between scavenger composition and scavenging efficiency was analyzed to understand the mechanisms of H₂S interaction with the scavengers. This evaluation is crucial for determining the real-world applicability of the scavengers in various operational settings.

The testing procedure was prepared based on the following solutions for the order of chemical interaction:

- Flask 1: Formation of H₂S, 200 ml of deionized water + 0.8780g of FeS.
- Flask 2: 2% solution of the sequestrant, 200 ml of the solution in deionized water.
- Flask 3: 200 ml of a 3% (w/v) solution of CdSO₄ in deionized water.
- Flask 4: 200 ml of a 3% (w/v) solution of CdSO₄ in deionized water.
- Kitassato: 200 ml of a 1N NaOH solution to neutralize any H₂S remains that may have passed through the system.

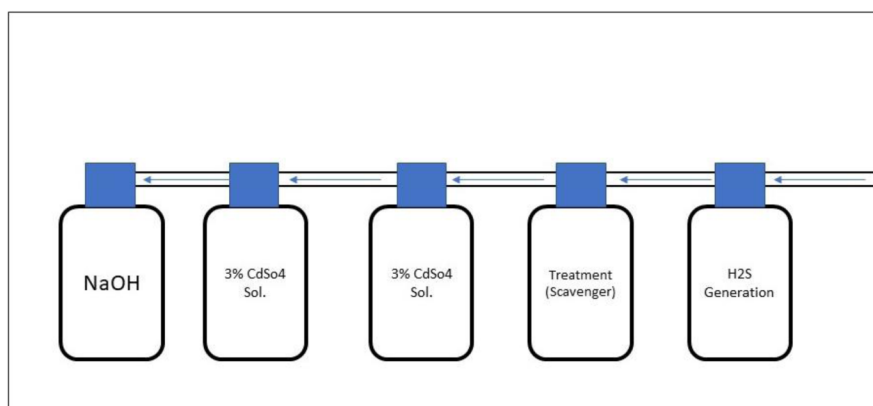


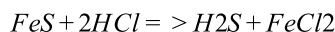
Figure 1—sequence of treatment of H₂S reaction.

To establish a functional system, it is essential to assemble the components, including the gas washer bottles, while ensuring a secure and leak-free connection, and then agitate flasks 1 and 2 to enhance the solubility of their contents. Next, prepare flask 1 by attaching the addition funnel and introducing 50ml of hydrochloric acid (HCl PA) while keeping the funnel closed to prevent premature formation of hydrogen sulfide (H₂S). The nitrogen system should then be connected, maintaining a low flow rate of approximately 7.5 L/min to avoid excessive pressure buildup, and once the system is verified to be leak-free, the nitrogen inlet should be closed to allow the pressure to stabilize. Throughout the setup and initialization process, it is mandatory to wear a gas mask to prevent inhalation of hazardous fumes, and extreme caution should be exercised when adding HCl, as it may cause liquid to flow back into the first flasks or release gas through the funnel, with all procedures being conducted in accordance with the guidelines outlined in the table to ensure a safe and controlled environment.

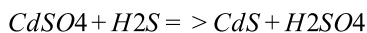
Table 1—HCl addition sequence.

HCl PA.	Time after the test (min)
10 ml	0
+ 10 ml	15
+ 20 ml	25
+10 ml	40

Reaction in Flask 1:



Reaction in Flask 3 + 4:



The resultant residue, comprising Cadmium Sulfide (CdS), shall serve as a quantitative indicator of the amount of hydrogen sulfide (H₂S) that was not effectively captured by the scavenger. To facilitate accurate analysis of this residue, a standardized protocol shall be employed. Initially, the residue shall be filtered utilizing a Whatman 5 filter to remove any impurities. Subsequently, the filtered residue shall be subjected to drying at a temperature of 105°C for a duration of one hour. The dried residue shall then be weighed to obtain an initial mass measurement. To ensure that the residue has reached a state of constant weight, the weighing process shall be repeated after 2 and 3 hours of drying, respectively. Finally, the obtained values shall be compared with the results from a blank sample and a market standard to verify the accuracy and reliability of the measurement, thereby ensuring the validity of the data obtained from the analysis of the CdS indicator.

$$\% = \frac{(\text{Blank Test Residue} - \text{H}_2\text{S Scavenger Residue}) * 100}{\text{Blank Test Residue}}$$

Flash Point Measurements. Flash points were determined following ASTM D93 standards to evaluate the flammability risk associated with each scavenger. The flash point is the lowest temperature at which the vapors of a volatile material will ignite when given an ignition source. This assessment is critical for safe transportation and storage of the scavengers, as highlighted in prior investigations (Thompson et al., 2020). Understanding the flash point of each scavenger allows operators to implement appropriate safety measures during handling.

Specific Gravity Determinations. The specific gravity of each scavenger was measured using a pycnometer to assess density variations. This property can influence the effectiveness and handling of the material and has been previously correlated with scavenging performance in related studies. The specific gravity measurements provide insights into the concentration of active ingredients in the scavenger formulations, which can directly impact their performance. To determine the specific gravity of the sample at various temperatures, we use a dilution method where the sample is mixed with a base fluid at different concentrations, ranging from 100% sample (90% zinc content) to 0% sample (100% base fluid).

Data Analysis

The data generated from each analytical technique were compiled and systematically analyzed to identify trends and correlations. Statistical methods, including regression analysis and ANOVA, were employed to determine the significance of the findings, ensuring robust and reliable conclusions. The integration of data from multiple analytical techniques allowed for a comprehensive assessment of each scavenger's performance and quality.

Results

Evaporation Studies

The results of the evaporation studies indicated significant variations in the thermal stability of the samples. For example, sample A exhibited a rapid evaporation rate, leading to residual impurities that compromised its effectiveness. In contrast, sample B showed a slower evaporation rate with minimal residue, indicating higher thermal stability and potential suitability for field applications.

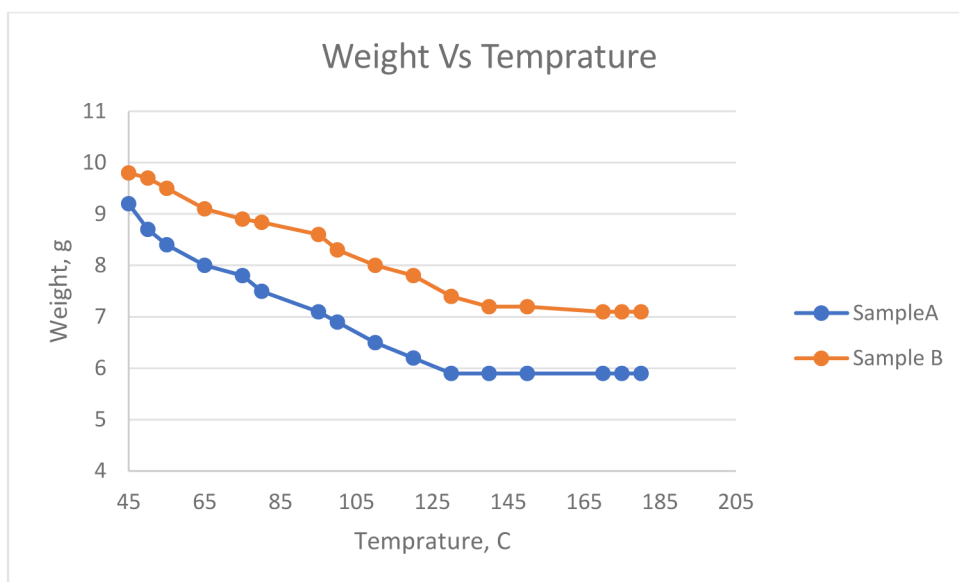


Figure 2—Evaporation rate of Scavenger samples.

Nuclear Magnetic Resonance (NMR) Spectroscopy

The NMR analysis demonstrated distinct differences in molecular structure among the samples, which correlated with their respective synthesis conditions. Notably, Sample B exhibited a unique chemical shift, indicating the presence of additional functional groups that may contribute to enhanced scavenging capabilities. This observation is consistent with previous literature, which highlights the significant influence of molecular structure on scavenging efficiency.

The samples were prepared by dilution in deuterated chloroform (CDCl_3). NMR spectra were acquired using a JEOL 500 MHz NMR spectrometer equipped with a 5 mm liquid probe. Both ^1H and $^{13}\text{C}\{^1\text{H}\}$ spectra were recorded using standard parameters. Figures 3 and 4 present the stacked ^1H and $^{13}\text{C}\{^1\text{H}\}$ NMR spectra of the active ingredient and the formulation, respectively.

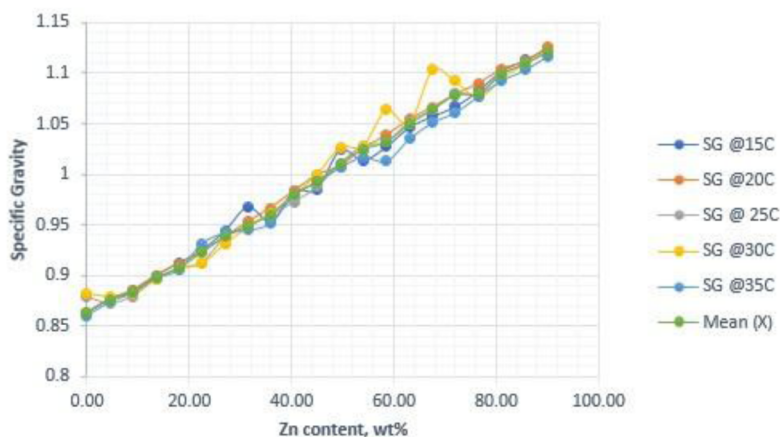


Figure 3—Specific gravity Trend against the Zinc Content in H₂S scavenger in different temperatures.

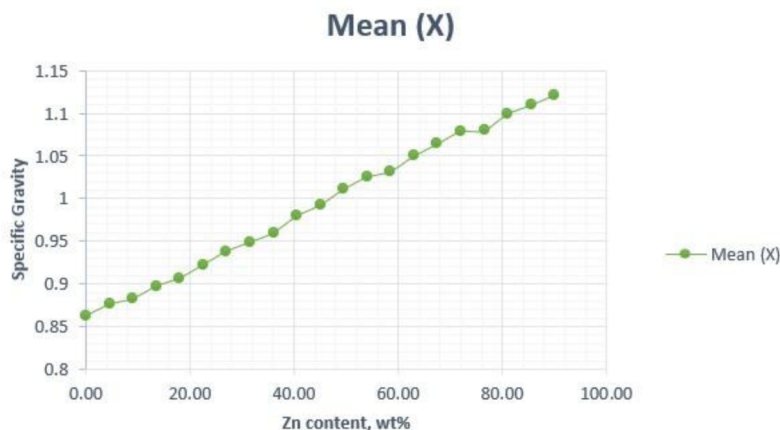


Figure 4—Mean specific gravity Trend against the Zinc Content in H₂S scavenger in different temperatures.

Given that the solvent in the formulation was aliphatic in nature, it was expected that no aromatic peaks would be observed in its spectra. As evident from the ¹H NMR spectra, aliphatic peaks at 2.1 ppm and in the range of 3.25–4.5 ppm were present in the formulation but absent in the active ingredient. This suggests that the solvent was effectively removed from the formulation. This observation was further corroborated by the ¹³C{¹H} NMR spectrum, where minor aliphatic peaks observed in the formulation were not detected in the active ingredient.

Liquid-state NMR spectroscopy was applied to a formulation and its isolated product. It was found that the isolation attempt of the active ingredient was relatively successful, evidenced by the removal of the formulation solvent peaks from the sample.

Scavenging Capability Evaluations

Scavenging capability evaluations demonstrated a robust relationship between scavenger composition and H₂S removal efficiency. Sample B achieved a remarkable 94%% H₂S removal efficiency within a specific timeframe, while sample A only managed 50%. These results underscore the importance of precise formulation and highlight the need for rigorous quality control measures during production. From left to right respectively: Blank Test, sample A and sample B. It can be seen in the Table below that the scavenger product showed no apparent CdS residue in any of the flasks, with the sample remaining slightly yellow only in Sample 3.

Table 2—Scavenging efficiency of both samples.

Test ref.	Time to form bottom residue (min)	CdS residue Filtration (g)	Efficiency (%)
Blank test	25	0.4969	-
SAMPLE A	25	0.2483	50.03
SAMPLE B	Did not form residue	0.0303	93.88

Sample B has a high-performance formulation designed to effectively capture H₂S and maintain its integrity across various seasonal conditions without degrading.

Flash Point Measurements

Flash point measurements varied significantly across the samples, with sample A exhibiting a flash point of 60°C, indicating potential risks in transportation. In contrast, Sample B had a flash point of 90°C, suggesting

a safer profile for handling. These findings emphasize the importance of flash point assessments in ensuring compliance with safety regulations.

Table 3—Flash point Results for both samples.

Sample	Flash Point °C
A	60
B	114

Specific Gravity Determinations

Specific gravity assessments indicated notable density variations among the samples. Sample B, with a specific gravity of 1.13 g/cm³, was found to be more effective in H₂S scavenging compared to sample A, which had a specific gravity of 1.0 g/cm³. Previous studies have correlated density with scavenging performance, suggesting that higher density may enhance the interaction between the scavenger and H₂S molecules. It can be seen in [Figure 3](#) and [Figure 4](#) represent the effect zinc wt.% on the specific gravity at different temperatures starting from the base fluid.

Discussion

The findings of this study reinforce the necessity for comprehensive evaluation protocols for zinc-based H₂S scavengers. The combined methodologies employed herein not only facilitate the detection of impurities but also contribute to a deeper understanding of the factors influencing scavenger performance.

Comparative Analysis of Results

To further understand the implications of the results, a comparative analysis was conducted among the various samples. This analysis enabled the identification of trends in performance relative to the chemical composition and structural attributes observed via NMR.

Evaporation Rates and Impurities

The evaporation studies highlighted that samples with higher residual impurities significantly affected their H₂S scavenging capabilities. For instance, samples with a high concentration of volatile organic compounds (VOCs) tended to exhibit lower scavenging efficiencies.

NMR Spectroscopy Insights

NMR spectroscopy revealed that samples containing specific functional groups, such as thiol (-SH) or amine (-NH₂), showed enhanced reactivity with H₂S, leading to improved scavenging performance. This finding aligns with the theoretical understanding of how these functional groups interact with H₂S to form stable complexes.

Scavenging Efficiency and Chemical Composition

The correlation between chemical composition and scavenging efficiency was further underscored by the scavenging capability evaluations. For example, samples with a higher concentration of zinc ions exhibited superior performance in H₂S removal, confirming the critical role of zinc as an active scavenging agent.

Flash Points and Safety Considerations

The flash point measurements provided essential insights into the safety profiles of the samples. Samples with lower flash points posed significant risks during transportation and storage, necessitating the implementation of stringent safety protocols.

Specific Gravity and Performance Correlation

Finally, the specific gravity assessments revealed that higher density samples not only performed better in scavenging but also demonstrated greater thermal stability during evaporation studies. This correlation suggests that the formulation of scavengers should consider density as a vital parameter for optimizing performance.

Implications for Industry Standards

The correlations identified between molecular structure, thermal stability, and scavenging efficiency underscore the complexity of scavenger formulations and the need for meticulous quality assurance processes. The current industry practices may benefit from the incorporation of the multi-faceted evaluation approach demonstrated in this study, which could lead to standardized testing protocols that enhance safety and efficiency in drilling operations.

Integration of Analytical Techniques

This research also highlights the importance of integrating various analytical techniques. While traditional methods may provide limited insight into scavenger quality, the multi-faceted approach adopted in this study offers a more thorough assessment, setting a new standard for the industry. By employing NMR spectroscopy and other analytical methods, the study was able to uncover molecular insights that may have otherwise remained undetected, paving the way for improved formulations and enhanced operational safety.

Safety Considerations

Furthermore, the implications of flash point and specific gravity measurements underscore the importance of safety in the handling and transportation of scavengers. The discrepancies in flash points among the tested samples illustrate the critical need for safety evaluations as part of the quality assessment process. Ensuring that operators are equipped with scavengers that possess favorable safety profiles can significantly mitigate risks associated with H₂S exposure.

Future Research Directions

Future research should focus on refining these methodologies and exploring additional scavenger formulations to further enhance safety standards in the industry. The ongoing development of new materials and formulations presents an opportunity to improve the efficiency and safety of H₂S scavengers. Additionally, continuous monitoring and evaluation of scavenger performance in real-world applications will be vital to ensuring ongoing effectiveness and safety.

Conclusion

This study presents a pioneering approach to the evaluation of zinc-based H₂S scavengers, establishing a new benchmark for quality assessment in the industry. The integration of multiple analytical techniques enables the identification of previously undetectable impurities and defects, thereby enhancing the overall safety and efficiency of drilling operations. The findings underscore the necessity of rigorous testing protocols to ensure the reliability and quality of H₂S scavengers.

As the oil and gas industry continues to prioritize safety, adopting comprehensive evaluation methodologies is essential for stakeholders involved in the procurement and application of H₂S scavengers. The insights gained from this study can serve as a foundation for future developments in scavenger technology and operational safety.

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